ALICE L WALTON SCHOOL OF MEDICINE

von Willebrand Disease

Last updated October 28, 2025 14 min read ScholarRx

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Adapted from: von Willebrand Disease

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of von Willebrand disease (vWD).
2. Describe the pathophysiology of vWD, including its inheritance pattern.
3. Interpret clinical findings to properly diagnose vWD.
4. Outline the management plan of vWD.

CASE CONNECTION

AJ is a 32-year-old female who comes to your clinic with concerns about heavy menses. "My period has always been super heavy. My mom and sister say the same thing, but no one ever told them why." When you ask if she has noted any other unusual bleeding, AJ stops and thinks. "No, not really. Except for the nosebleeds." AJ has two aunts who have similar symptoms. AJ's exam, including her gynecologic exam, is normal. Her vitals are within normal limits.

What tests will be helpful in diagnosing AJ's condition? How will you explain her condition to her? Consider your answers as you read, and we'll revisit AJ at the end of the brick.

GO TO CONCLUSION ↓

What Is von Willebrand disease?

von Willebrand disease (vWD) is a bleeding disorder caused by a deficiency or a defect in von Willebrand factor (vWF), a blood clotting protein crucial to platelet function. vWF is the factor involved in the initial formation of the platelet plug, which binds platelets to subendothelial collagen to stop bleeding. Many patients have mild symptoms, and some have no symptoms at all.

vWD is the most common inherited bleeding disorder and is most prevalent in people of European descent.

Why do patients with vWD have difficulty in stopping bleeding?

How Do Patients With von Willebrand Disease Present (LO #1)?

Severity of vWD varies widely, and many patients go undiagnosed because they have no clinically significant symptoms. Many patients are not diagnosed until they are subjected to a significant hemostatic challenge (surgery, dental extraction, major trauma).

When symptoms are present, they often are mild and non–life threatening. They include:

- Easy bruising
- Bleeding from mucous membranes (frequent nosebleeds)
- Excessive bleeding from wounds
- Menorrhagia (he

vWD is more often diagnosed in women because they are more likely to notice prolonged bleeding during menstruation or childbirth.

Why might many cases of vWD go undiagnosed?

What Is the Pathophysiology von Willebrand Disease (LO #2)?

vWD has three main types and a number of subtypes. Classification is based on the amount of vWF, size of the multimers, and multimer functionality. We won't go into these details in this discussion but will focus on the most characteristic genetic and functional problems.

Genetics

There are several ger are inherited in an **autosomal dominant pattern** (Figure 1). However, some cases can be autosomal recessive or occur via acquired random mutations.

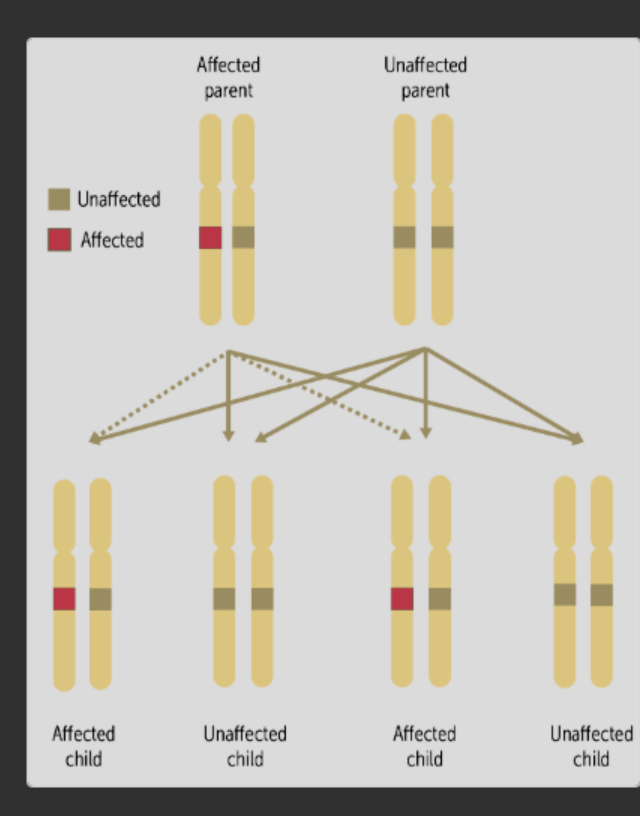

Figure 1

Because of the dominant inheritance pattern of vWD, most people afflicted with the disease are heterozygous for the gene defect. Very rarely, homozygous deficiencies arise and generate more severe symptoms, sometimes resulting in clinically significant factor VIII deficiencies, causing symptoms resembling hemophilia.

Normal Function of von Willebrand Factor

vWD is due to defective vWF. We'll review the molecule's normal function first, then talk about what goes wrong in the disease.

vWF is a glycoprotein involved in blood clotting. vWF is multimeric, meaning it is built from multiple molecules (i.e., monomers) held together with noncovalent bonds; its subunits contain binding sites for both platelet glycoprotein receptors and collagen. During the clotting process, platelets aggregate and many clotting factors are activated to form fibrin. When vWF is deficient or does not function properly, blood clotting is impaired.

vWF has two major functions in coagulation:

- It binds platelets to the subendothelium in a process called **platelet adhesion**.
- It binds and stabilizes **factor VIII** in the circulation.

Let's start with **platelet adhesion**. When blood vessels are damaged, platelets must be recruited to slow and stop blood leakage into the surrounding tissue. In damaged vessels, endothelial damage leads to exposed sub-endothelium, and the collagen that resides there is exposed to the circulating blood. vWF acts as the glue that adheres platelets to this exposed collagen via platelet glycoprotein Ib (GpIb) receptors (Figure 2). This process of adhesion is the first step in building the platelet plug, which serves as a stopgap to prevent blood loss through the damaged vessel.

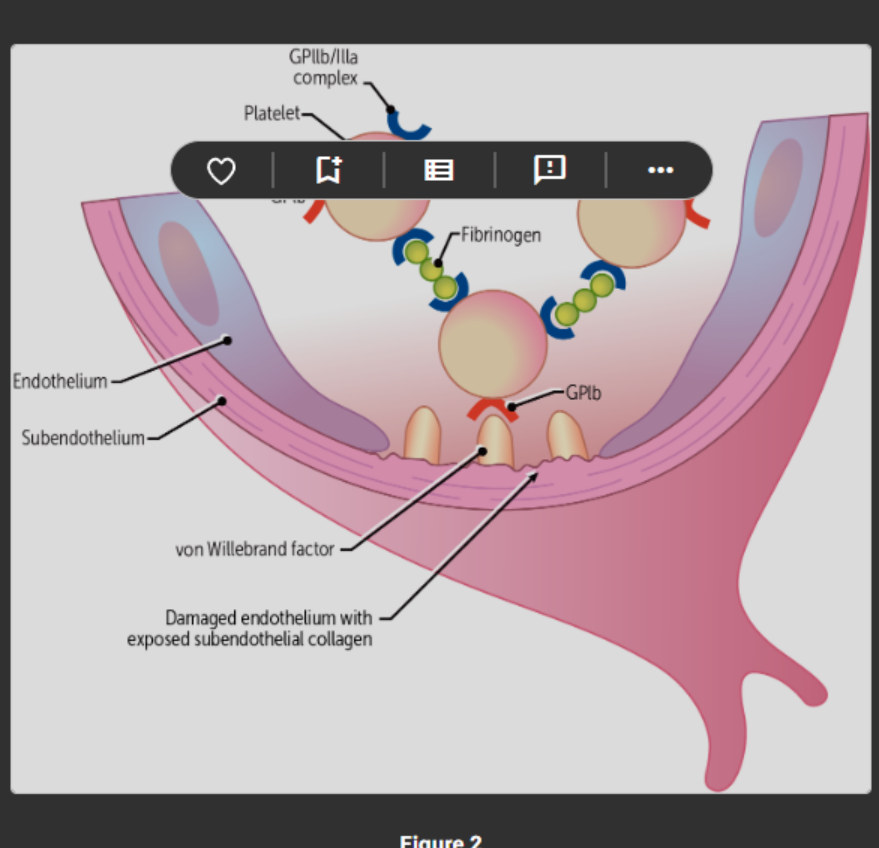

Figure 2

The second job of vWF is to **bind to and stabilize factor VIII.** Factor VIII is a coagulation factor that rapidly degrades unless it is bound to vWF. We can think of vWF as factor VIII's protector; vWF will bind circulating factor VIII and protect it from degradation until it is needed in the coagulation cascade.

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Mechanism of Disease

In most cases of vWD, genetic abnormalities in vWF directly cause **impaired platelet adhesion** and reduced platelet plug formation.

In addition to this primary mechanism, the defects in vWF can lead to inadequate protection of factor VIII against degradation. This reduces factor VIII levels, leaving insufficient amounts for coagulation. In a small number of cases, the decrease in available factor VIII is so pronounced that the disease mimics hemophilia A (a hereditary disease in which factor VIII is decreased or absent).

CLINICAL CORRELATION

A thrombotic (clotting) disease that is the "opposite" of vWD is thrombotic thrombocytopenic purpura (TTP). In this disease there is impaired cleavage of ultra-large vWF multimers. The result is excess vWF and widespread platelet aggregation (causing thrombosis) and eventually depletion of circulating platelets (causing bleeding).

Variants

Von Willebrand disease has three main types, aptly named Types 1, 2, and 3. Here are some general principles to understand about these three types:

- Type 1 (mildest and most common form): this is a partial absence of von Willebrand factor, but the function is intact. This type is inherited in an autosomal dominant fashion
- Type 2 (qualitative abnormality): the amount of von Willebrand factor may be normal or decreased, but it is functionally abnormal. There are multiple subtypes of type 2 that is beyond the scope of this discussion. It is predominantly inherited in an autosomal dominant fashion.
- Type 3 (most severe form): complete absence of von Willebrand factor, inherited in an autosomal recessive fashion. These patients also have decreased Factor VIII, so they often have primary and secondary hemostasis bleeding.

CLINICAL REASONING

Remember, types 1 and 3 are quantitative abnormalities (normal vWF function but partial or complete loss of it), type 2 is a qualitative abnormality (may have normal amount of vWF, but it is functionally abnormal).

How Do We Diagnose von Willebrand Disease (LO #3)?

As previously noted, ue to lack of symptoms. Evaluation focuses on the patient's personal and family history, followed by laboratory studies to rule out more common causes of bleeding, and then tests to confirm vWD.

History and Physical Examination

The history will include a family history looking for bleeding disorders. The physical exam should include a search for bruises or signs of bleeding, especially in the gums and mouth. Petechiae (tiny dots of bleeding within the skin and mucosal surfaces) may be seen. Less often, the stools may be heme positive for blood.

Laboratory Tests

A variety of tests may be needed to diagnose vWD, including vWF antigen and activity tests as well as other bleeding disorder tests.

Quantitative tests used to measure the quantity, size, and function of vWF include:

- **vWF antigen:** a von Willebrand factor antigen assay may be used to measure the amount of vWF in the blood.
- **Ristocetin cofactor and vWF activity tests:** function of vWF can be measured using the antibiotic ristocetin, which activates vWF and binds platelets, causing agglutination in the test tube.

Additional tests that are less specific include:

- **Platelet aggregation studies:** used to see how well the platelets are functioning.
- **Partial thrombo** ess the intrinsic arm of the coagulation cascade. If factor VIII is decreased, the PTT will be prolonged.
- **Factor VIII clotting activity:** measures the amount of factor VIII in the blood in order to rule out the possibility of factor VIII deficiency, an alternative diagnosis for von Willebrand disease.

How does ristocetin cause platelet agglutination?

CLINICAL CORRELATION

People with blood type O generally have lower levels of vWF even without any VWD gene mutations. People with blood type O and low vWF (even vWF levels within the range of type 1 vWD) but no bleeding symptoms should **not** be diagnosed with von Willebrand disease.

How Do We Manage von Willebrand Disease (LO #4)?

The overall goal of vWD treatment is bleeding control. Specific treatment depends on the severity and site of bleed and type of vWD. Most patients respond to **vWF replacement**.

The treatment options include:

- **vWF replacement therapies:** direct supplementation of vWF is used to increase the circulating levels of vWF. This is especially valuable in vWD patients who require rapid correction of major bleeding, or prior to surgery. The recombinant version is approved for use only in adults. Children may be treated with plasma- derived concentrates, which also contain factor VIII.
- **Concentrate alternatives:** when vWF concentrates are not available, fresh frozen plasma and cryoprecipitate are **last** choice, alternative sources of vWF for major bleeding. Fresh frozen plasma is plasma frozen after collection of whole blood or plasma from donors, while cryoprecipitate is a concentrated solution of clotting factors obtained from freezing and thawing fresh frozen plasma.

CLINICAL CORRELATION

These alternatives are not preferred as they are not intensively treated to prevent infection. Additionally, FFP does not contain much vWF, so large volumes would be required for any effect.

- **Desmopressin (DDAVP):** this drug releases vWF from cellular compartments and is useful in patients with residual endothelial storage of vWF. Not all vWD patients have such stores, so DDAVP does not work in all patients and is most often used in less severe bleeding.
- **Antifibrinolytic drugs:** aminocaproic acid and tranexamic acid work by preventing dissolution of the formed clots. They are useful alongside DDAVP in moderate bleeding, or in preventing mucosal bleeding during surgical procedures (e.g., dental procedures).
- **Platelet transfusion:** in certain types of vWD platelets are used to correct thrombocytopenia or platelet aggregation defects in patients with acquired or inherited platelet disorders.

Depending on the severity of bleed and baseline vWF levels, patients might need therapy for up to 14 days.

What is the role of desmopressin in treating patients with von Willebrand disease (vWD)?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to AJ, what tests are helpful to the diagnosis? How do you explain her condition to her?

AJ's history and family history of bleeding suggest a hereditary bleeding disorder. You weigh the risk of vWD vs other, less common bleeding disorders. Her complete blood count (including platelet count) results are normal. You reason that if vWD is more common, you should start your evaluation there. You do a vWF antigen assay and a ristocetin platelet agglutination test, which both come back abnormal, showing a decrease in vWF quantity and function. You explain to AJ that her excessive menstrual bleeding and nosebleeds are caused by the inability of platelets to anchor to the site of an injured blood vessel. "Maybe that's what my mom and sister have, too," AJ says. "Are my children at risk?" You explain that vWD is an inherited disease; she has a 50% chance of passing the condition to her children.

Summary

- von Willebrand disease (vWD) is the most common inherited bleeding disorder, often presenting with symptoms such as excessive bleeding, easy bruising, and heavy menstruation.
- vWD is caused by a deficiency or defect in von Willebrand factor (vWF), a protein essential for blood clotting and platelet function.
- vWF serves two m atelets to the subendothelium (tor VIII in the bloodstream.
- Diagnosis of vWD involves a series of laboratory tests that evaluate the quantity, size, and function of vWF, as well as tests to rule out other bleeding disorders.
- Treatment for vWD depends on the severity and location of bleeding, as well as the specific type of vWD. Options include desmopressin, vWF replacement therapies, antifibrinolytic drugs, and platelet transfusion.
- Despite its prevalence, many cases of vWD go undiagnosed due to mild or absent symptoms.

Review Questions

1. The loss of von Willebrand factor in von Willebrand disease (vWD) will most likely result in increased degradation of which of the following?

A. ○ Endothelial cells
B. ○ Factor VII
C. ● Factor VIII
D. ○ Factor IX
E. ○ Platelets

Hide Explanation

The correct answer is factor VIII (C). One major function of von Willebrand factor (vWF) is to bind and stabilize circulating factor VIII until it is needed in the coagulation cascade. If vWF is severely deficient, factor VIII levels will be low, a situation that can mimic hemophilia A (which is characterized by a decrease in, or absence of, factor VIII). vWF is involved in anchoring platelets to endothelial cells; its loss would not result in the degradation of either endothelial cells (A) or platelets (E). Factor VII (B) and factor IX (D) are other clotting factors that do not interact with vWF.

2. A mother has been diagnosed with vWD and is found to be heterozygous for the disease. She is having a child with a father who has no history of blood clotting disorders. What are the chances that their child will also have the disease?

A. ○ 0%
B. ○ 25%
C. ● 50%
D. ○ 100%
E. ○ Cannot be determined

Hide Explanation

The correct answer is attern for von Willebrand disease (vWD) is autosomal dominant, even though there are certain types that can be inherited in an autosomal recessive pattern or arise sporadically. Because of this autosomal dominant inheritance pattern, most affected people are heterozygous for the gene defect, and it is rare to see someone who is homozygous. Given that the father has no signs of the disease, we can assume he is homozygous for normal von Willebrand factor (vWF), and the mother is heterozygous for the gene defect. This would give the child a 50% chance of inheriting the disease. Therefore, 0% (A), 25% (B), 100% (D), and cannot be determined (E) would all be incorrect.

3. Which of the following treatments for von Willebrand disease (vWD) contains bioactive von Willebrand factor (vWF)?

A. ○ Aminocaproic acid
B. ● Cryoprecipitate
C. ○ Desmopressin
D. ○ Platelet transfusion
E. ○ Tranexamic acid

Hide Explanation

The correct answer is cryoprecipitate (B). Cryoprecipitate is a concentrate of clotting factors including von Willebrand factor (vWF) made from serum. It is used as an alternative source for and disease (vWD) when vWF concentrat s do not contain vWF. Aminocaproic acid (A) and tranexamic acid (E) are antifibrinolytics that prevent dissolution of formed clots and may help prevent mucosal bleed in patients with vWD, but are less used in major bleeding. Desmopressin (C) releases vWF from cellular compartments and is useful in patients with residual endothelial storage of vWF, but is less used in major bleeding compared to vWF concentrates. Platelet transfusion (D) is used to correct thrombocytopenia or platelet aggregation defects, which may be one of the causes for abnormal coagulation despite normal vWF levels.

4. A 17 year-old male is referred to your Emergency Room for refractory bleeding which started immediately after a wisdom tooth was extracted and which the dentist was unable to stop. Physical examination is unremarkable except for bleeding from the back of his mouth. He notes that he has always had nosebleeds about once per week which take up to two hours to stop. His father and sister also have frequent nosebleeds. PT/PTT are normal. CBC is normal, including a normal platelet count and platelet appearance on blood smear. Which of the following is the most likely diagnosis?

A. ○ Bernard-Soulier syndrome
B. ○ Hemophilia A
C. ○ Hemophilia B
D. ● von Willebrand disease

Hide Explanation

The correct answer is von Willebrand disease (D). The diagnosis would be confirmed with VWF antigen and activity testing. Bernard-Soulier syndrome (A) is an inherited platelet disorder, caused by a mutation in Glycoprotein Ib (VWF receptor), that presents with decreased platelets (thrombocytopenia) that are abnormally large in appearance. Hemophilia A (B) and Hemophilia B (C) are both X-linked disorders (would not occur in both males and females) that present with deeper tissue bleeding (muscle bleeds, bleeding into joints). They would also both have elevated PTT levels.

How would you rate this Brick?

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.

Check off the objectives when you feel comfortable with the concept

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of von Willebrand disease (vWD).~~
02 ☑ ~~Describe the pathophysiology of vWD, including its inheritance pattern.~~
03 ☑ ~~Interpret clinical findings to properly diagnose vWD.~~
04 ☑ ~~Outline the management plan of vWD.~~

Objective Confidence: 4/4